120 mm

270 mm

1105345 DS152C/5

10420

For the use only of Registered Medical Practitioner or a Hospital or a Laboratory.

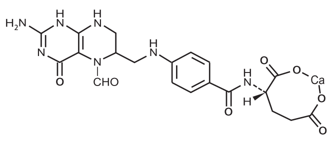
CAFONATE

LEUCOVORIN CALCIUM INJECTION USP

COMPOSITION
Each ml contains,
Leucovorin Calcium USP
Equivalent to Leucovorin 10 mg

PRODUCT DESCRIPTION
A clear, yellowish solution.
After dilution: A clear, yellowish solution.

DESCRIPTION
Leucovorin Calcium Injection is a sterile clear, yellowish solution. It contains sodium chloride, disodium edetate, water for injection as excipients.
Leucovorin is one of several active, chemically reduced derivatives of folic acid. It is useful as an antidote to drugs which act as folic acid antagonists. Also known as folinic acid, Citrovorum factor, or 5-formyl-5,6,7,8-tetrahydrofolic acid, this compound has the chemical designation of Calcium N - [p-[[[(6RS) - 2 - amino - 5 - formyl - 5,6,7,8- tetrahydro - 4 - hydroxy - 6 - pteridinyl]methyl]amino]benzoyl] -L -glutamate (1:1). Molecular formula of Leucovorin Calcium is $C_{20}H_{21}CaN_7O_7$ and molecular weight is 511.50 The structural formula of Leucovorin calcium is:



CLINICAL PHARMACOLOGY
Pharmacodynamics
Pharmacotherapeutic group: Detoxifying agents for antineoplastic treatment; ATC code: V03AF03
Leucovorin calcium is the calcium salt of 5-formyl tetrahydrofolic acid. It is an active metabolite of folinic acid and an essential coenzyme for nucleic acid synthesis in cytotoxic therapy.
Leucovorin calcium is frequently used to diminish the toxicity and counteract the action of folate antagonists, such as methotrexate. Leucovorin calcium and folate antagonists share the same membrane transport carrier and compete for transport into cells, stimulating folate antagonist efflux. It also protects cells from the effects of folate antagonist by repletion of the reduce folate pool. Leucovorin calcium serves as a pre-reduced source of H4 folate; it can therefore bypass folate antagonist blockage and provide a source for the various coenzyme forms of folic acid.
Leucovorin calcium is also frequently used in the biochemical modulation of fluoropyridine (5-fluorouracil) to enhance its cytotoxic activity. 5-fluorouracil inhibits thymidylate synthase (TS), a key enzyme involved in pyrimidine biosynthesis, and Leucovorin calcium enhances TS inhibition by increasing the intracellular folate pool, thus stabilising the 5- fluorouracil-thymidylate synthase complex and increasing activity.
Finally, intravenous Leucovorin calcium can be administered for the prevention and treatment of folate deficiency when it cannot be prevented or corrected by the administration of folic acid by the oral route. This may be the case during total parenteral nutrition and severe malabsorption disorders. It is also indicated for the treatment of megaloblastic anaemia due to folic acid deficiency, when oral administration is not feasible.

Pharmacokinetics
Absorption
Following intramuscular administration of the aqueous solution, systemic availability is comparable to an intravenous administration. However, lower peak serum levels (Cmax) are achieved.
Distribution
The distribution volume of folinic acid is not known. Peak serum levels of the parent substance (D/L-5-formyl-tetrahydrofolic acid, folinic acid) are reached 10 minutes after i.v. administration and 28 minutes after intramuscular administration.
AUC for L-5-formyl-THF and 5-methyl-THF were 28.4 ± 3.5 mg.min/l and 129 ± 112 mg.min/l after a dose of 25 mg. The inactive D-isomer is present in higher concentration than L-5-formyltetrahydrofolate. Folate is concentrated in the cerebrospinal fluid, although distribution occurs to all body tissues.
Biotransformation
Leucovorin calcium is a racemate where the L-form (L-5-formyl-tetrahydrofolate, L-5-formyl- THF), is the active enantiomer. The major metabolic product of folinic acid is 5-methyltetrahydrofolic acid (5-methyl-THF) which is predominantly produced in the liver and tetrahydrofolic acid (5-methyl-THF) which is predominantly produced in the liver and intestinal mucosa.
Peak levels of 5-methyl-THF are observed at 1.3 and 2.8 hours following intravenous and intramuscular administration, respectively. The terminal half-life for total reduced folates is reported as 6.4 hours.
Elimination
The elimination half-life is 32 - 35 minutes for the active L-form and 352 - 485 minutes for the inactive D-form, respectively.
The total terminal half-life of the active metabolites is about 6 hours (after intravenous and intramuscular administration).
80-90 % with the urine (5- and 10-formyl-tetrahydrofolates inactive metabolites), 5-8 % with the faeces.

INDICATIONS
Leucovorin calcium is indicated
a) to diminish the toxicity and counteract the action of folic acid antagonists such as methotrexate in cytotoxic therapy and overdose in adults and children. In cytotoxic therapy, this procedure is commonly known as "Calcium Folinate Rescue"
b) in combination with 5-fluorouracil in cytotoxic therapy.

CONTRAINDICATIONS
- Hypersensitivity to the active substance or to any of the products excipients.
- Pernicious anaemia or other anaemias due to vitamin B12 deficiency.

DOSAGE & ADMINISTRATION
Calcium Folinate Rescue in methotrexate therapy:
Since the Calcium Folinate Rescue dosage regimen depends heavily on the posology and method of the intermediate- or high-dose methotrexate administration, the methotrexate protocol will dictate the dosage regimen of Calcium Folinate Rescue. Therefore, it is best to refer to the applied intermediate or high dose methotrexate protocol for posology and method of administration of Leucovorin calcium.
The following guidelines may serve as an illustration of regimens used in adults, elderly and children:
Calcium Folinate Rescue has to be performed by parenteral administration in patients with malabsorption syndromes or other gastrointestinal disorders where enteral absorption is not assured. Dosages above 25-50 mg should be given parenterally due to saturable enteral absorption of Leucovorin calcium.
Calcium Folinate Rescue is necessary when methotrexate is given at doses exceeding 500 mg/m^2 body surface and should be considered with doses of 100 mg – 500 mg/m^2 body surface.
Dosage and duration of Calcium Folinate Rescue primarily depend on the type and dosage of methotrexate therapy, the occurrence of toxicity symptoms, and the individual excretion capacity for methotrexate. As a rule, the first dose of Leucovorin calcium is 15 mg (6-12 mg/m^2) to be given 12-24 hours (24 hours at the latest) after the beginning of methotrexate infusion. The same dose is given every 6 hours throughout a period of 72 hours. After several parenteral doses treatment can be switched over to the oral form however in the presence of gastrointestinal toxicity, nausea, or vomiting, Leucovorin calcium should be administered parenterally.
In addition to Leucovorin calcium administration, measures to ensure the prompt excretion of methotrexate (maintenance of high urine output and alkalinisation of urine) are integral parts of the Calcium Folinate Rescue treatment. Renal function should be monitored through daily measurements of serum creatinine.
Forty-eight hours after the start of the methotrexate infusion, the residual methotrexate-level should be measured. If the residual methotrexate-level is >0.5 μmol/l, Leucovorin calcium dosages should be adapted according to the following table:.

Residual methotrexate blood level 48 hours after the start of the methotrexate administration:	Additional Leucovorin calcium to be administered every 6 hours for 48 hours or until levels of methotrexate are lower than 0.05 μmol/l:
> 0.5 μmol/l	15 mg/m^2
> 1.0 μmol/l	100 mg/m^2
> 2.0 μmol/l	200 mg/m^2

Delayed methotrexate excretion may be seen in some patients. This may be caused by a third space accumulation (as seen in ascites or pleural effusion for example), renal insufficiency or inadequate hydration. Under such circumstances, higher doses of Leucovorin calcium and/or prolonged administration may be indicated.
In combination with 5-fluorouracil in cytotoxic therapy:
Different regimens and different dosages are used, without any dosage having been proven to be the optimal one.
The following regimens have been used in adults and elderly in the treatment of advanced or metastatic colorectal cancer and are given as examples. There are no data on the use of these combinations in children. Particular care should be taken when treating elderly or debilitated patients as these patients are at increased risk of severe toxicity with this therapy.
Bimonthly regimen: Leucovorin calcium 200 mg/m^2 by intravenous infusion over two hours, followed by bolus 400 mg/m^2 of 5-FU and 22-hour infusion of 5-FU (600 mg/m^2) for 2 consecutive days, every 2 weeks on days 1 and 2.
Weekly regimen: Leucovorin calcium 20 mg/m by bolus i.v. injection or 200 to 500 mg/m^2 as i.v. infusion over a period of 2 hours plus 500 mg/m^2 5-fluorouracil as i.v. bolus injection in the middle or at the end of the Leucovorin calcium infusion.
Monthly regimen: Leucovorin calcium 20 mg/m^2 by bolus i.v. injection or 200 to 500 mg/m^2 as i.v. infusion over a period of 2 hours immediately followed by 425 or 370 mg/m 5- fluorouracil as i.v. bolus injection during five consecutive days.
For the combination therapy with 5-fluorouracil, modification of the 5-fluorouracil dosage and the treatment-free interval may be necessary depending on patient condition, clinical response and dose limiting toxicity as stated in the product information of 5-fluorouracil. A reduction of Leucovorin calcium dosage is not required.
The number of repeat cycles used is at the discretion of the clinician.
Antidote to the folic acid antagonists trimetrexate, trimethoprim, and pyrimethamine:

Trimetrexate toxicity:
- Prevention: Leucovorin calcium should be administered every day during treatment with trimetrexate and for 72 hours after the last dose of trimetrexate. Leucovorin calcium can be administered either by the intravenous route at a dose of 20 mg/m^2 for 5 to 10 minutes every 6 hours for a total daily dose of 80 mg/m^2, or by oral route with four doses of 20 mg/m^2 administered at equal time intervals. Daily doses of Leucovorin calcium should be adjusted depending on the haematological toxicity of trimetrexate.
- Overdosage (possibly occurring with trimetrexate doses above 90 mg/m^2 without concomitant administration of Leucovorin calcium): after stopping trimetrexate, Leucovorin calcium 40 mg/m^2 IV every 6 hours for 3 days.

Trimethoprim toxicity:
- After stopping trimethoprim, 3-10 mg/day Leucovorin calcium until recovery of a normal blood count.

Pyrimethamine toxicity:
- In case of high dose pyrimethamine or prolonged treatment with low doses, Leucovorin calcium 5 to 50 mg/day should be simultaneously administered, based on the results of the peripheral blood counts.

Method of administration
For intravenous and intramuscular administration only. In the case of intravenous administration, no more than 160 mg of Leucovorin calcium should be injected per minute due to the calcium content of the solution.
For intravenous infusion, Leucovorin calcium may be diluted with 0.9% sodium chloride solution or 5% glucose solution before use.
Leucovorin calcium must not be injected intrathecally.

Instructions for use
For intravenous infusion, Leucovorin calcium may be diluted with sodium chloride 9 mg/ml (0.9%) solution for injection or glucose 50 mg/ml (5%) solution for injection before use.
Prior to administration, Leucovorin calcium should be inspected visually. The

Cafonate DS (Malaysia Export)
Unfold Size: 120 x 270 mm **(Front)**
Folds: (VF-VF) Zig Zag - VF - VF
Folded Size: 120 x 22.5mm
Paper: 40 GSM Bible paper
Color: Black Text on White Paper
Artwork Code: DS152C/5
Item Code: 1105345
Pharma Code: 10420

Naprod Packaging Development:

Artwork prepared by: Tejan Surve	**PP No.:** INJ015292	
Artwork Approval Date:	**PP Date:** 20/01/2022	
Prepared by Artcell Dept.	**Checked by PD Dept.**	**Reviewed & Approved by QA Dept.**

Artwork History: Change Control No.:
1. In point Product Description text 'After dilution: A clear colourless to yellowish solution' changed to 'After dilution: A clear, yellowish solution'.
2. Date of revision changed from 'July 2024"' to '14 February 2026'.
3. Pharma code changed from 8442 to 10420
4. Manufacturing Address Updated.
5. Text changed 'Jauhi dari kanak-kanak' to 'Jauhkan daripada capaian kanak-kanak.'
Artwork code changed from DS152C/4 to DS152C/5.

120 mm

Blank space behind Pharma Code

270 mm

solution for injection or infusion should be a clear and yellowish solution. If cloudy in appearance or particles are observed, the solution should be discarded.
Leucovorin calcium solution for injection or infusion is intended only for single use. Any unused medicinal product or waste material should be disposed of in accordance with the local requirements.

OVERDOSAGE
There have been no reported sequelae in patients who have received significantly more Leucovorin calcium than the recommended dosage. However, excessive amounts of Leucovorin calcium may nullify the chemotherapeutic effect of folic acid antagonists.
Should overdosage of the combination of 5-fluorouracil and Leucovorin calcium occur, the overdosage instructions for 5-fluorouracil should be followed.

ADVERSE REACTIONS
Frequencies are defined using the following convention:
Very common ($\geq$1/10); Common ($\geq$1/100 to $<$1/10); Uncommon ($\geq$1/1,000 to $<$1/100); Rare ($\geq$1/10,000 to $<$1/1,000); Very rare ($<$1/10,000); Frequency not known (cannot be estimated from the available data).
Monotherapy:

System Organ Class	Frequency	Adverse Effect
Immune system disorders	Very rare	Hypersensitivity, Anaphylactoid reaction, Anaphylactic reaction
	Frequency Not known	Anaphylactic shock
Psychiatric disorders	Rare	Insomnia, Agitation and Depression[a]
Gastrointestinal disorders	Rare	Gastrointestinal Disorders[a]
Nervous system disorders	Rare	Seizure[b]
	Frequency Not known	Syncope
Skin and subcutaneous tissue disorders	Very rare	Urticaria
General disorders and administration site conditions	Uncommon	Pyrexia[c]

a: After high doses
b: Increase in the frequency of attacks in epileptics
c: Has been observed after administration of Leucovorin calcium as solution for injection
Combination therapy with 5-fluorouracil only:
Generally, the safety profile depends on the applied regimen of 5-fluorouracil due to enhancement of the 5-fluorouracil induced toxicities. Additional undesirable effects when used in combination with 5-fluorouracil are presented in below table:

System Organ Class	Frequency	Adverse Effect
Blood and lymphatic system disorders	Very common	Bone marrow failure,[m] Leukopenia, Neutropenia, Thrombocytopenia, Anaemia
Gastrointestinal disorders	Very common	Vomiting and Nausea[n]. Diarrhoea and Dehydration[o] Stomatitis
Metabolism and Nutrition disorders	Frequency not known	Hyperammonaemia
Skin and subcutaneous Tissue disorders	Common	Palmar-plantar erythrodysaesthesia syndrome[p]
General disorders and administration site conditions	Very common	Mucosal inflammation[m], Cheilitis.

m: Including fatal cases
n: Monthly regimen: No enhancement of other 5-fluorouracil induced toxicities (e.g. neurotoxicity)
o: Weekly regimen: With higher grades of toxicity and dehydration, resulting in hospital admission for treatment and even death
p: Hand-foot syndrome

WARNINGS
Leucovorin calcium should only be given by intramuscular or intravenous injection and must not be administered intrathecally.
When folinic acid has been administered intrathecally following intrathecal overdose of methotrexate death has been reported.

General
Leucovorin calcium should be used with methotrexate or 5-fluorouracil only under the direct supervision of a clinician experienced in the use of cancer chemotherapeutic agents.
Leucovorin calcium treatment may mask pernicious anaemia and other anaemias resulting from vitamin B12 deficiency.
Many cytotoxic medicinal products – direct or indirect DNA synthesis inhibitors – lead to macrocytosis (hydroxycarbamide, cytarabine, mecaptopurine, thioguanine). Such macrocytosis should not be treated with folinic acid.
In epileptic patients treated with phenobarbital, phenytoin, primidone, and succinimides there is a risk to increase the frequency of seizures due to a decrease of plasma concentrations of anti-epileptic drugs. Clinical monitoring, possibly monitoring of the plasma concentrations and, if necessary, dose adaptation of the anti-epileptic drug during Leucovorin calcium administration and after discontinuation is recommended.

Leucovorin calcium/5-fluorouracil
Leucovorin calcium may enhance the toxicity risk of 5-fluorouracil, particularly in elderly or debilitated patients. The most common manifestations are leucopenia, mucositis, stomatitis and/or diarrhoea, which may be dose limiting. When Leucovorin calcium and 5- fluorouracil are used in combination, the 5-fluorouracil dosage has to be reduced more in cases of toxicity than when 5-fluorouracil is used alone.
Combined 5-fluorouracil/Leucovorin calcium treatment should neither be initiated nor maintained in patients with symptoms of gastrointestinal toxicity, regardless of the severity, until all of these symptoms have completely disappeared.
Because diarrhoea may be a sign of gastrointestinal toxicity, patients presenting with diarrhoea must be carefully monitored until the symptoms have disappeared
completely, since a rapid clinical deterioration leading to death can occur. If diarrhoea and/or stomatitis occur, it is advisable to reduce the dose of 5-fluorouracil until symptoms have fully disappeared. Especially the elderly and patients with a low physical performance due to their illness are prone to these toxicities. Therefore, particular care should be taken when treating these patients.
In elderly patients and patients who have undergone preliminary radiotherapy, it is recommended to begin with a reduced dosage of 5-fluorouracil.
Leucovorin calcium must not be mixed with 5-fluorouracil in the same IV injection or infusion.
Calcium levels should be monitored in patients receiving combined 5-fluorouracil/Leucovorin calcium treatment and calcium supplementation should be provided if calcium levels are low.

Leucovorin calcium/methotrexate
Leucovorin calcium has no effect on non-haematological toxicities of methotrexate such as the nephrotoxicity resulting from methotrexate and/or metabolite precipitation in the kidney. Patients who experience delayed early methotrexate elimination are likely to develop reversible renal failure and all toxicities associated with methotrexate. The presence of pre-existing or methotrexate-induced renal insufficiency is potentially associated with delayed excretion of methotrexate and may increase the need for higher doses or more prolonged use of Leucovorin calcium.
Excessive Leucovorin calcium doses must be avoided since this might impair the antitumour activity of methotrexate, especially in CNS tumours where Leucovorin calcium accumulates after repeated courses.
Resistance to methotrexate as a result of decreased membrane transport implies also resistance to folinic acid rescue as both medicinal products share the same transport system.
An accidental overdose with a folate antagonist, such as methotrexate, should be treated as a medical emergency. As the time interval between methotrexate administration and Calcium Folinate Rescue increases, Leucovorin calcium effectiveness in counteracting toxicity decreases.
Laboratory tests
The possibility that the patient is taking other medications that interact with methotrexate (e.g. medications which may interfere with methotrexate elimination or binding to serum albumin) should always be considered when laboratory abnormalities or clinical toxicities are observed.
The following provides general advice for monitoring patients; however, specific monitoring recommendations may vary with local medical practice.
Leucovorin calcium/5-fluorouracil
Full blood count (FBC) with differential and platelets: prior to each treatment, weekly during the first two courses, and at the time of anticipated white blood cell (WBC) nadir in all courses thereafter.
Electrolytes and liver function tests: prior to each treatment for the first three courses and prior to every other course thereafter.
Leucovorin calcium/methotrexate
Serum creatinine levels and serum methotrexate levels: at least once daily.
Urine pH: in cases of methotrexate overdose or delayed excretion, monitor as appropriate, to ensure maintenance of pH 7.0.

Fertility, Pregnancy and Lactation
Pregnancy:
There are no adequate and well-controlled clinical studies conducted in pregnant or breast-feeding women. Animal studies are insufficient with respect to reproductive toxicity. However, there are no indications that folic acid induces harmful effects if administered during pregnancy. During pregnancy, 5-flurouracil and methotrexate should only be administered on strict indications, where the benefits of the drug to the mother should be weighed against possible hazards to the foetus.
Should treatment with methotrexate or other folate antagonists take place despite pregnancy or lactation, there are no limitations as to the use of Leucovorin calcium to diminish toxicity or counteract the effects.
5-fluorouracil use is generally contraindicated during pregnancy and contraindicated during breastfeeding; this applies also to the combined use of Leucovorin calcium with 5- fluorouracil.

Lactation:
It is not known whether this drug is excreted in human milk.
Leucovorin calcium can be used during breast-feeding when considered necessary according to the therapeutic indications.

Fertility
Leucovorin calcium is an intermediate product in the metabolism of folic acid and occurs naturally in the body. No fertility studies have been conducted with Leucovorin calcium in animals.

DRUG INTERACTIONS
When Leucovorin calcium is given in conjunction with a folic acid antagonist (e.g. cotrimoxazole, pyrimethamine, antibiotic with antifolic effect, methotrexate) the efficacy of the folic acid antagonist may either be reduced or completely neutralised.
Leucovorin calcium may diminish the effect of anti-epileptic substances: phenobarbital, primidone, phenytoin and succinimides, and may increase the frequency of seizures (a decrease of plasma levels of enzymatic inductor anticonvulsant drugs may be observed because the hepatic metabolism is increased as folates are one of the cofactors).
Concomitant administration of Leucovorin calcium with 5-fluorouracil has been shown to enhance the efficacy and toxicity of 5-fluorouracil.
Concurrent administration of chloramphenicol and folic acid in folate deficient patients may result in antagonism of haematopoietic response to folic acid.

Effects on ability to drive and use machines
There is no evidence that Leucovorin calcium has an effect on the ability to drive or use machines.

INCOMPATIBILITIES
Incompatibilities have been reported between injectable forms of Leucovorin calcium and injectable forms of droperidol, 5-fluorouracil, foscarnet and methotrexate.

STORAGE
Store between 2°C and 8°C. Protect from light.
Keep out of reach of children. Jauhkan daripada capaian kanak-kanak.

Storage condition for Infusion:
Store below 8°C. Protect from light. Keep out of reach of children. Jauhkan daripada capaian kanak-kanak.

SHELF LIFE
24 Months

Shelf life for Infusion:
3 days (72 hours)

DOSAGE FORMS AND PACKAGING AVAILABLE
CAFONATE (Leucovorin Calcium Injection USP) is available in a vial containing Leucovorin Calcium USP equivalent to Leucovorin 50mg/5ml & 300mg/30ml.

Packing: 1 vial/box, 10vials/box. No other than these pack sizes are marketed.

Date of revision: February 2026.

Product Registration Holder:
Pahang Pharmacy Sdn. Bhd.,
Lot 5979, Jalan Teratai,
5½ Miles Off Jalan Meru,
41050 Klang Selangor, MALAYSIA

Manufactured in India by:
NAPROD
Naprod Life Sciences Pvt. Ltd.
Plot No. G-17/1, MIDC, Tarapur, Boisar,
Dist. Palghar 401506 Maharashtra State,
India.

120 x 270mm

Cafonate DS (Malaysia Export)
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